Dreaming of Windows

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The National Fenestration Rating Council (NFRC) provides labels such as this describing the performance of windows.

Q We are building a new passive-solar home and have had difficulty finding commercially available low-U, high solar heat gain coefficient (SHGC) windows. The specs quoted for the windows in the *Home Energy* article "The Little House That Could" sound like a dream.

Could you please provide more information on how to go about finding windows with these properties? We have not seen windows with similar specs through Pella, Marvin, Loewen, Jeld-Wen, and so on. (Jeld-Wen even is listed in Green Building Products as having high SHGC low-e coatings, but the specs on its windows do not seem to reflect this.) Do major manufacturers not offer this type of glazing? Or do we need to request expensive custom windows?

Looking for Windows That Perform
Pittsburgh, Pennsylvania

A. "The Little House That Could," (*HE* Nov/Dec '06, p. 24) presents a shining example of how high solar gain low-e windows are used with properly sized overhangs to minimize heating loads in a zero-energy home. All low-e coatings reduce temperature-driven heat transfer through double or multiple glazings, but different types of low-e coatings influence the amount of solar heat gain transmitted through the glass. The vast majority of low-e windows sold today are low solar gain products, designed to minimize air conditioning and limit swing season overheating. While low solar gain low-e products are suitable for a one-size-fits-all approach to specifying windows (they minimize most discomfort issues in all climates), there are opportunities for high solar gain products. These include homes in northern climates without cooling loads; homes that incorporate overhangs and other summer season solar-control methods, such as trees and awnings; homes in northern climates with small window areas; and passive-solar homes.

Unfortunately, high solar gain products are hard to find. Local manufacturers should be asked for low-e products with a National Fenestration Rating Council (NFRC) solar heat gain coefficient (SHGC) of approximately 0.45 or greater. Sometimes high solar gain products are available as special-order items, but local salespeople may not know this. What follows is some guidance on how to order such products from major national manufacturers. This list is not inclusive; customers interested in windows from other manufacturers should pursue this topic with the manufacturers' representatives and not give up after the first salesperson says, "What?"

Marvin Windows offers an option for a high solar gain low-e coated product. This option is a pyrolitic coating with excellent visible and solar transmittance; its somewhat higher emittance means that the U-factor of the window will be about 0.03 (10%) higher than that of a typical sputtered low-e coating, but the SHGC will be about 65% higher. While Marvin does not advertise this option, the customer should ask for a hard-coat low-e coating. It is available with Marvin's wood and clad-wood lines, but not with the fiberglass Integrity line. If a local Marvin dealer does not recognize this option, customers should call Marvin at (800)727-7742 and ask for a service representative.

Pella also offers products with a high solar gain low-e coating. These products are limited to the manufacturer's Designer Series; they utilize the low-e coating on a low-e hinged glass panel (HGP), formerly known as a double glazing panel (DGP). These products are available in most window operator types and in glass doors; they come with aluminum-clad wood frames. The Designer Series also offers products that come with integral shades, which, if used effectively, can improve U-factors and provide for solar control. Customers who want integral shades should ask for either

- a Designer 2 product (double glazed) with a low-e HGP; or
- a Designer 3 product (triple glazed) with a clear insulating glass unit (IG) and a low-e HGP.

While the Pella Web site offers general information on the Designer Series, I recommend visiting a showroom and talking with a knowledgeable salesperson.

Andersen offers, as a special-order product intended for passive-solar designs, a high solar gain low-e window. Andersen recommends appropriate design strategies to filter out summer sun and prevent swing season overheating. This variant of the company's regular low-e is manufactured using the same sputtering process as Andersen uses for its standard HP (high performance) products for consistent coloration and view clarity. HP 4 has about 85% of the solar gain of a clear-glass window, with a U-factor close to that of standard HP. To order this product, the customer will have to find an Andersen dealer familiar with the special-ordering process and ask for Original HP 4 (also known as Cardinal Low-e 178) glass. The special-ordering process may entail an extra charge and a longer wait. If needed, Andersen's customer service phone number is (800)426-4261.

Loewen Windows provides both double-glazed and triple-glazed windows with low-e coatings. While Loewen's standard product line is low solar gain low-e, the company does sell high solar gain low-e glazing systems. To order these systems, it is best to call a Loewen Window Center or a Loewen Authorized Dealer. A list of dealers can be found at www.Loewen.com.

Over a dozen manufacturers in the United States and Canada offer window products with Southwall Technologies Heat Mirror low-e coated polyester film. In these products, the low-e coated thin film is stretched between two pieces of glass. Southwall offers a variety of coatings, from high solar gain to low solar gain. Because of the added air or gas space, these products tend to have lower U-factors than standard low-e products. The high solar gain products (HM88 and HM TC88) are offered by Kensington, Alpen, Newtown-Slocomb, Regency, Republic, Traco, Wellington, Stanek, Gilkey, High Performance Glass, Trans America Glass, and Paradigm. See the Dealer Locator section of the Southwall Web site for specific contact information: www.southwall.com/southwall/Home/ResourceLibrary/DealerLocator/XIR/HeatMirror.html. Note that this list includes both window manufacturers and Heat Mirror insulating glass fabricators (who sell to window manufacturers).